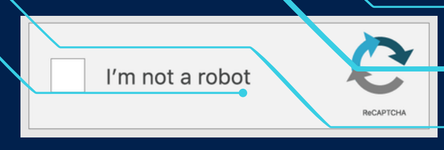
DETECTGPT

DETECTING AND INTERPRETING TEXT GENERATED BY LARGE LANGUAGE MODELS



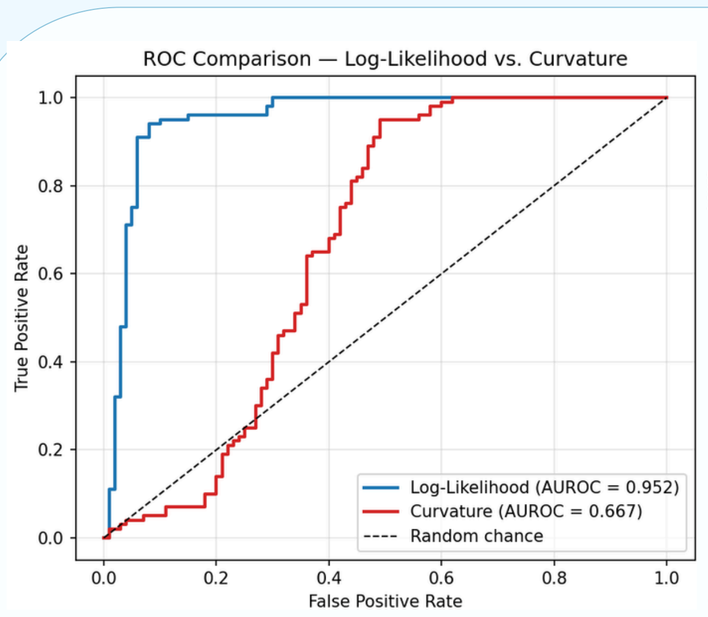
Large language models can generate text that closely resembles human writing, making reliable detection difficult. This project investigates different approaches to detecting machine-generated text, while also considering how detection can work across languages and how the evidence behind a prediction can be made visible.

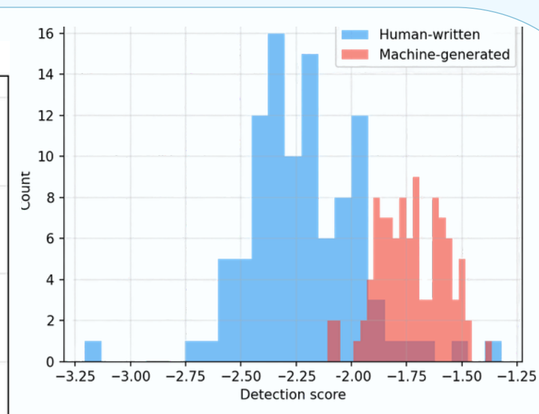
STATISTICAL DETECTION

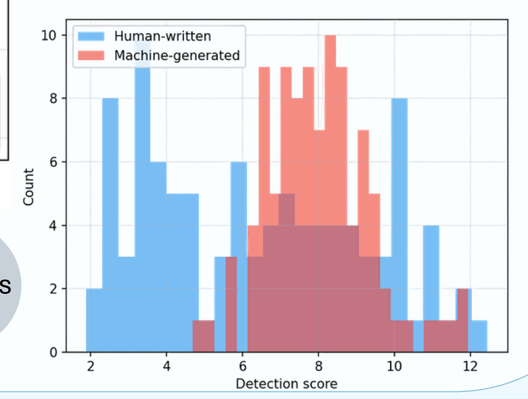
Question / objective Compare log-likelihood and curvature-based (DetectGPT) scoring under identical conditions, same dataset, same generator, same scoring backbone to isolate detector method as the only variable.

Method 500 paired XSum/Gemma 3 4B articles. Llama 3.2 3B scoring backbone. Thresholds calibrated via Youden's J on validation only. Curvature: 30 T5-perturbations per sample, standardised discrepancy (DetectGPT formula).

Takeaway Under matched conditions, log-likelihood scored higher (AUROC 0.952 vs 0.667 but this reflects performance in THIS controlled setup, not a general claimthat log-likelihood beats curvature. Curvature's theory assumes scorer ≈ generator; here they differ, which likely explains the gap more than any flaw in the method itself.







Log-likelihood's curve dominates curvature's across nearly all thresholds, with heavy overlap in curvature's score distributions explaining its lower AUROC.

INTERPRETABILITY

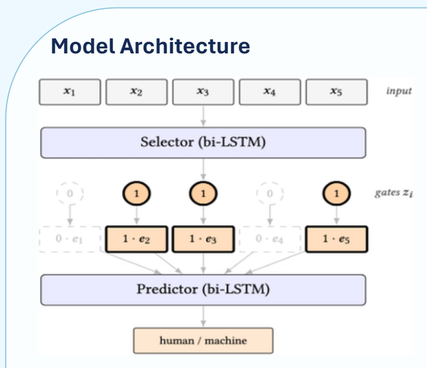
Aim Test whether a detector can still perform well when it only uses the parts of the text it selects as evidence, and whether showing that evidence helps people judge whether text is human- or machine-generated.

Method We train a BiLSTM text classifier on 8,160 balanced human- and machine-written English passages. The rationale model selects about 40% of the visible text and predicts using only that selected evidence. We compare it with the same BiLSTM using the full text and a simpler word- and character-based classifier.

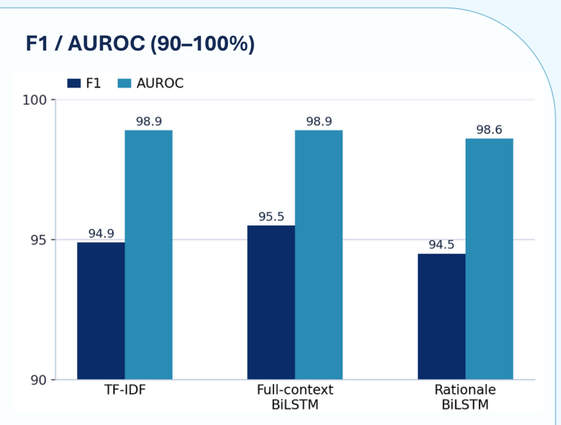
Results Using ~40% of the visible text, the rationale BiLSTM achieved 94.5% F1 and 98.6% AUROC, compared with 95.5% F1 and 98.9% AUROC with full context. In the human study, the highlighted rationales could not be shown to improve authorship judgements.

Concl. Strong detection remained possible using restricted evidence. Useful model evidence did not clearly improve human authorship judgements.

Model Architecture



F1 / AUROC (90–100%)



Rationale example (machine → machine)

The title refers to an improved Directivity Analysis method which is a variant of Mutual Coupling Based Linear Array (MCBLA). This technique enhances performance estimation in direction-of-arrival (DoA) measurement that utilizes a digital steerable antenna. It's particularly effective at suppressing sidelobe noise, improving tracking accuracy, and detecting weak signals compared to other conventional approaches. A series of simulations was carried out on the MCBLA with two signal sources over an infinite plane with free space propagation in order to study its impact on clutter rejection and DoA estimates. Overall findings conclude that the proposed technique has enhanced accuracy, robustness and reliability as compared to preeminent methods and may have potential applications for automotive and military systems.

FINE-TUNED MULTILINGUAL DETECTION

Objective Determine whether adapting models to isiXhosa and isiZulu using language-adaptive pretraining improves detection of LLM-generated text and what signals indicate it.

Method We create a dataset of 11,718 human/machine text pairs using FineWeb-2 for the human text and Gemma 3 4B for the continuations. This is used to evaluate mBERT and XLM-R with and without language-adaptive pretraining vs a TF-IDF baseline. We found the word signals that indicate LLM-generated text by looking at the TF-IDF word weights and built a classifier on three word groups and how often they appear.

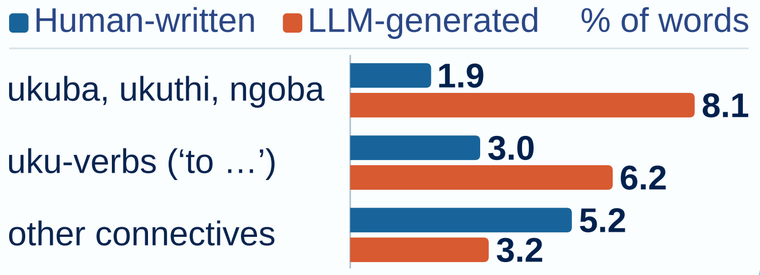
Results We found that none of the adapted vs unadapted comparisons showed statistically significant results at the default threshold. We also found that a TF-IDF baseline did as well as the transformer detectors, showing that the words used in the text were enough of an indicator. This was also supported by a classifier built on three word rates alone achieving an accuracy of 88.92%.

Takeaway Near-perfect detectors; adaptation showed no measurable gain; word choice was a strong indicator.

Model accuracy

TF-IDF baseline	**0.9974**
mBERT & XLM-R	**0.9970–0.9996**

Words used to identify if text is human-written or LLM-generated



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